Listening for Engine Faults: Can a Pre-trained Audio Model Triage Fleet Vehicles?

Industry AI Challenge, Team 2

Jeffrey Gregory and Jake Van Slyke

2026-11-11

1. Business problem

Client. The maintenance manager of a regional delivery fleet (~200 light vehicles) and the service writers who schedule shop time.

Problem. Engine faults such as *knock* (uncontrolled combustion that can damage pistons and bearings) and *misfire* (a cylinder failing to fire, which wastes fuel and can damage the catalytic converter) are expensive when caught late. Drivers notice "a weird noise" inconsistently, and technicians have limited time to triage complaints.

Current process. Drivers report problems verbally; a technician listens and runs an OBD-II scan. The engine computer flags many misfires with trouble codes, but intermittent or early problems may not set a code, and knock is often only heard. Without AI, the alternative is more frequent scheduled inspections.

What would make AI useful. A 10-second phone recording, taken during a routine pre-trip check, that flags vehicles likely to need an engine inspection. It needs no new hardware, and its job is to *prioritise* inspections, not replace the mechanic. There is commercial precedent: a used-car auction company reported about 21 % fewer engine-related buyer disputes while a sound-based fault detector, reviewed by humans, was switched on [8].

Cost of wrong predictions. A *missed fault* (false negative) lets a damaged engine keep running: possible engine or catalytic-converter repairs costing thousands, plus downtime. A *false alarm* (false positive) costs one unnecessary inspection (~30 min of technician time), and frequent false alarms erode trust in the tool. Missed faults are far more costly, so we prioritise recall while capping false alarms.

2. Model, data, and adaptation

Model. The Audio Spectrogram Transformer (AST, 86 M parameters) [1], fine-tuned by its authors on AudioSet [2], a 2-million-clip collection of labelled YouTube sounds that includes engine and vehicle categories. We chose it because it is a strong, widely used general-purpose audio model that runs from the Hugging Face `transformers` library and accepts our 16 kHz audio directly. **Alternative:** BEATs [3], a self-supervised audio model of similar size; for on-device use, a much smaller model such as YAMNet would be the reasonable option.

Dataset. KweriMoto Engine Sounds [4]: 1,067 mono 16 kHz clips (≤ 10 s) labelled `healthy` (351), `engine_knock` (608), or `misfire` (108), with predefined splits that keep every clip from a recording in one split — train 721, validation 154, test 154 — plus a separate `real_test` set of 38 real knocks. We checked that recording groups, overlays and duplicate hashes do not cross splits. The dataset card describes research and educational use and notes differences in the original sources' licensing.

Limitations relative to the business use. This dataset is not a fleet's recordings, and its structure creates a serious risk that a model learns *where a clip came from* instead of *what is wrong with the engine*:

- All misfire clips come from a single source; all AudioSet clips are knocks.
- Half of the knock clips are **synthetic** — an AudioSet knock sound mixed over a healthy car recording.
- `real_test` contains only knocks, so it cannot measure false alarms.
- No information about vehicle, microphone position, or engine speed; labels are not mechanic-confirmed.

Adaptation. (a) *Feature extraction:* AST is frozen and each clip becomes a 768-number summary; a linear classifier (2,307 parameters) is trained on top. (b) *Partial fine-tuning:* the top two AST layers are also trained.

3. Success criteria (set before training)

	Choice	Why
Model metric	**Fault recall**: average recall over knock and misfire	Missed faults are the costly error
Guardrail	**Healthy false-alarm rate**: share of healthy clips flagged	Frequent false alarms erode trust
Business criterion	Catch ≥ **85 %** of faulty engines while flagging ≤ **20 %** of healthy ones	Useful as a triage aid
Operational constraint	**< 1 s per 10 s clip on a laptop CPU**	Runs on a shop laptop CPU

We used validation for model selection and reserved the test split for final evaluation.

4. Experiments

We compared a majority-class benchmark, a baseline, and three experiments on the same validation split (Table 1). For trained models, we kept the epoch with the lowest validation loss.

Table 1: Validation results. Benchmark = always predict *knock* (the most common class).

Run	Fault recall	Healthy false alarms	Misfire recall
Benchmark	0.50	1.00	0.00
Baseline	0.82	0.19	0.88
Exp 1	0.81	0.21	0.75
Exp 2	0.86	0.19	1.00
Exp 3	0.78	0.15	0.88

Baseline (frozen AST + linear head). Fault recall 0.82 with a 0.19 false-alarm rate — far above the benchmark, but just short of the 85 % target. Validation loss is lowest at epoch 8 and rises afterwards (Figure 1): with only 721 training clips, even a linear head overfits quickly, so early stopping matters.

Experiment 1 — partial fine-tuning. *Expected:* adapting the top layers would help separate a knock from surrounding engine noise. *Happened:* knock recall improved (0.76 → 0.86), but misfire recall fell (0.88 → 0.75), so fault recall did not improve (Section 6).

Experiment 2 — class-weighted loss. *Expected:* weighting the rare misfire class (76 training clips) more heavily would raise misfire recall. *Happened:* misfire recall reached 16/16 and fault recall 0.86 with no change in false alarms. It was the only configuration to meet both criteria at the default seed. *Learned:* the weights recovered two misfires and missed three additional knocks. *Stability:* class weighting improved validation fault recall in all ten paired runs (84.7% mean vs 76.5%). The weighted head met both targets in 2 of 10 seeds.

Experiment 3 — no synthetic knocks in training. *Expected:* if synthetic knocks teach the real knock sound, removing them should hurt real-knock recall. *Happened:* real-knock recall *rose* (0.84 → 0.93) while synthetic-knock recall fell (0.67 to 0.42). *Learned:* the overlays helped detect similar synthetic clips. Every real knock in this validation set comes from AudioSet, so the gain could reflect recording-source cues. Removing the overlays also changes class balance.

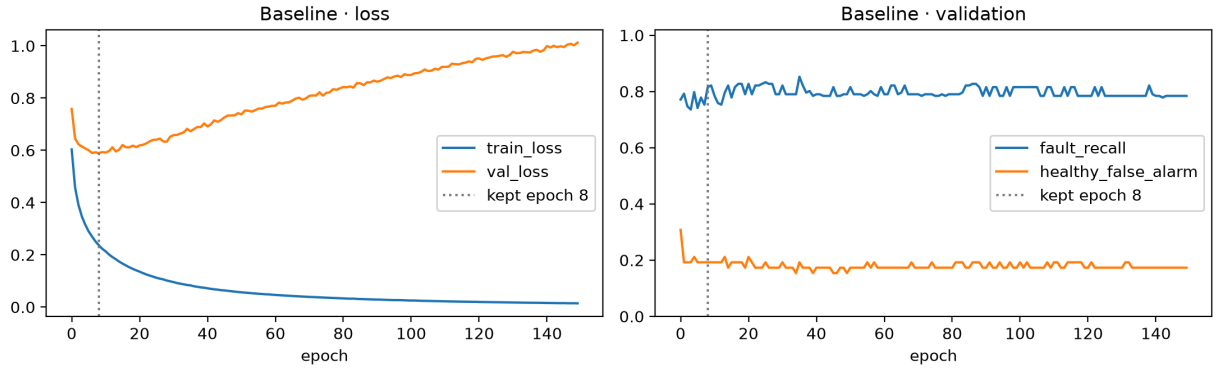
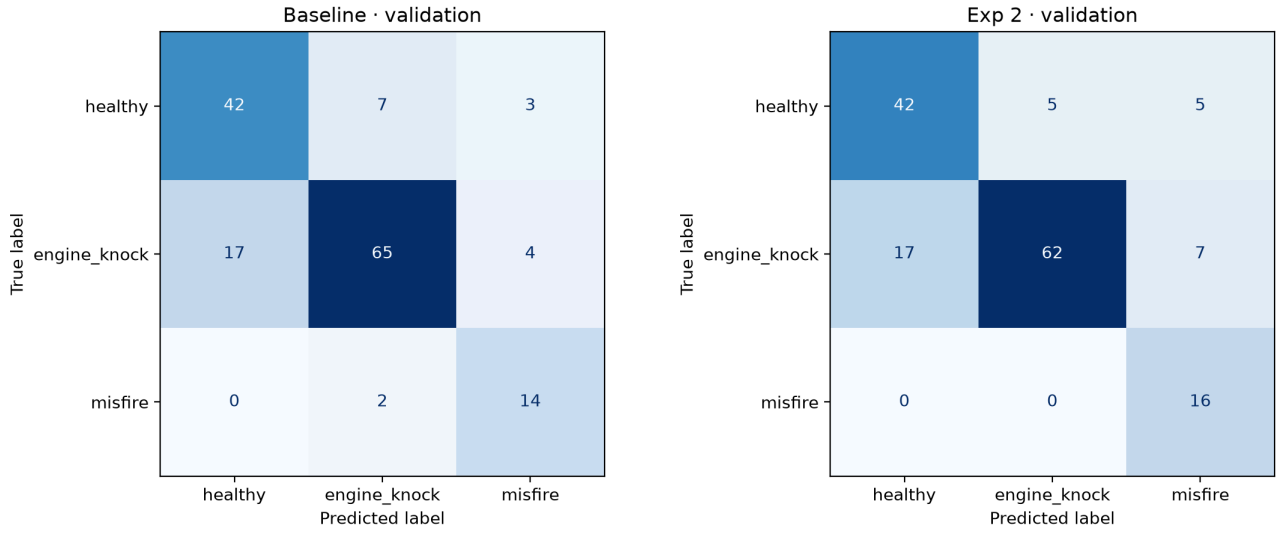


Figure 1: Baseline training curves. Left: loss; right: validation fault recall and false-alarm rate. The dotted line marks the epoch kept.

5. Evaluation

5.1. How well does it perform?

Experiment 2 reached 86.0% mean fault recall and flagged 19.2% of healthy clips on validation (Figure 2). The AST feature extraction step averaged 247 ms per clip on the test machine's CPU and 33 ms on its RTX 5060 Ti GPU. Timing on the shop's laptop remains to be measured.



(a) Baseline (b) Experiment 2

Figure 2: Validation confusion matrices (rows: true class; columns: predicted).

Final test (run once). We selected Experiment 2 on validation (higher mean fault recall across seeds). On the 154 held-out test clips it reached **fault recall 0.91** with a **0.11 false-alarm rate**: knock recall 0.81 (real 0.90, synthetic 0.72) and misfire 16/16. On `real_test` it caught 33 of 38 real knocks (0.87). The mixed test set meets both business criteria. The baseline scored close to the selected model (0.90 recall and 0.11 false alarms). The test set shares the training set's source structure, and overlap with AST's pretraining data is possible. Performance on independently recorded fleet vehicles remains open.

5.2. Where does it fail?

The baseline correctly predicted the misfire in `hf_misfire_011` with a 96.8% model score. Of 33 baseline errors, 13 were synthetic knocks predicted as healthy. Three cases show the costs of different errors. The proposed causes are hypotheses based on the clip metadata.

1. `revix_car_knocking_0043` — **knock predicted healthy with 97 % confidence.** A 4.1 s synthetic clip. Short knock clips (< 5 s) were misclassified 55% of the time (6 of 11) versus 20% for longer ones. A short recording may contain fewer audible knocks, and an overlay may be masked by engine noise. A missed knock could delay inspection.
2. `audioset_9wGJydlJvrk_30000` **(95.3% healthy).** This full-length real knock was also missed. Variation in microphone distance and background noise could contribute. A full-length recording offers no guarantee that the fault will be flagged.
3. `hf_healthy_059` **(92.2% misfire).** The clip was healthy. Normal variation in engine rhythm or recording conditions could contribute. A false alarm uses inspection time and repeated alarms can reduce trust.

The weighted head caught more faults on average while staying within the mean false-alarm limit. The examples show why technician judgment still matters even when the model score is high.

5.3. Is the model hearing the fault, or the source?

A linear probe on the same AST features identifies a clip's source with **95 %** accuracy (chance 54 %), and **98 %** among healthy clips alone. The notebook's recording-source diagnostics include three further checks.

Table 2: ROC AUC when the recording source is held fixed (0.5 = chance).

Model	Knock vs healthy (Revix only)	Misfire vs healthy (HF only)
Baseline	0.87	0.86
Exp 2 (class weights)	0.87	0.89
Exp 3 (no synthetic)	0.70	0.78

- **Within-source tests (Table 2):** comparing clips from the *same* source, the model still separates faulty from healthy (baseline AUC 0.865 for Revix knock and 0.859 for Hugging Face misfire). The Revix knocks are overlaid sounds and the misfire comparison has only 28 clips.
- **Zero-shot baselines:** CLAP [5], an audio–text model used with no training, scores below the benchmark (fault recall 0.22). AST's own "Engine knocking" output separates AudioSet knocks from healthy clips better than it separates Revix overlays from healthy clips (AUC 0.86 vs 0.62). Prior exposure to AudioSet or differences between the real and mixed audio could explain the gap.
- **Source-adversarial training:** we trained the representation to hide the source within each class [6]. The adversary was partly fooled, but a fresh probe still identified healthy-clip sources **perfectly** at every setting. That result shows the source signal survived the tested adversarial setup [7].

6. An unsuccessful experiment: partial fine-tuning

1. **Why we expected improvement.** AudioSet features are generic ("engine", "vehicle"); training the top layers should specialise them to engine faults, a standard way to improve transfer learning.
2. **What happened.** Fault recall stayed flat (0.82 → 0.81) and false alarms rose slightly (0.19 → 0.21), at ~6,000× more trainable parameters (14.2 M), minutes instead of seconds of training, and a 345 MB model to deploy.
3. **Evidence.** Validation loss bottomed out at epoch 2 while training loss trended down (Figure 3), a sign of rapid overfitting. Knock recall rose but misfire fell from 14/16 to 12/16: the model improved on the class with the most training data (399 knocks) at the expense of the smallest (76 misfires).
4. **What we learned.** On these 721 training clips, fine-tuning improved the majority class and reduced recall on the smallest class.
5. **Effect on our recommendation.** We retained the frozen backbone. It achieved slightly higher fault recall with much less training and a small classifier to update.

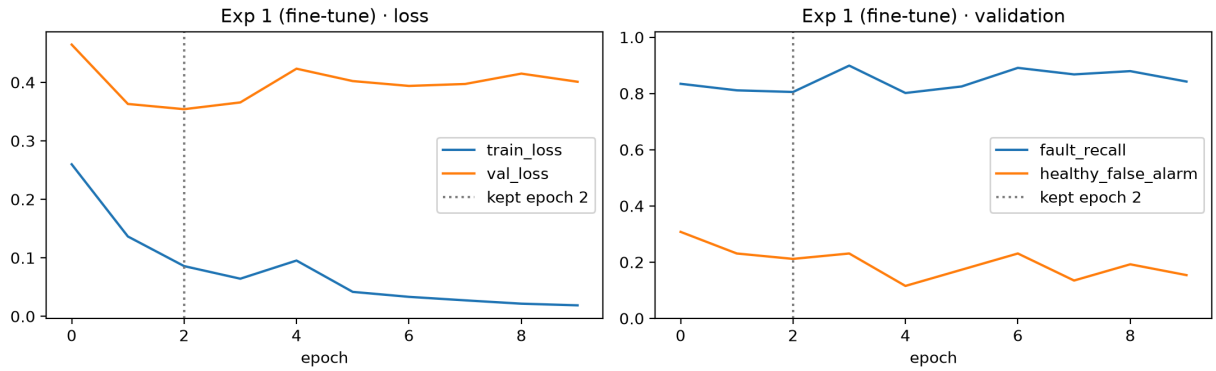


Figure 3: Fine-tuning curves. Validation loss reached its minimum at epoch 2.

7. Industry recommendation

Continue development before a limited pilot. The class-weighted frozen model met both targets on the mixed test set and ran within the CPU time budget on our test machine. Its within-source results justify further evaluation. Recording source remains easy to identify, all misfires come from one source, and only 2 of 10 validation seeds met both targets. The knock-only test split measures knock recall, while false alarms need to be measured on healthy fleet vehicles.

Collect healthy and faulty recordings across the fleet using the same phone placement and recording procedure [8]. Have technicians confirm labels using inspection findings and diagnostic codes. Hold out entire vehicles and measure mean fault recall, healthy false alarms and latency on shop hardware. A limited pilot would be justified if the fleet evaluation reaches at least 85% mean fault recall and at most 20% healthy false alarms, with inference under one second per clip.

8. References

1. Y. Gong, Y.-A. Chung, J. Glass. "AST: Audio Spectrogram Transformer." *Interspeech*, 2021.
2. J. F. Gemmeke et al. "Audio Set: An ontology and human-labeled dataset for audio events." *ICASSP*, 2017.
3. S. Chen et al. "BEATs: Audio Pre-Training with Acoustic Tokenizers." *ICML*, 2023.
4. KweriMoto Engine Sounds dataset. Hugging Face: `rayyyyy007/kwerimoto-engine-sounds`.
5. Y. Wu et al. "Large-scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation." *ICASSP*, 2023.
6. Y. Ganin et al. "Domain-Adversarial Training of Neural Networks." *JMLR*, 2016.
7. Y. Elazar, Y. Goldberg. "Adversarial Removal of Demographic Attributes from Text Data." *EMNLP*, 2018.
8. D. Fedorishin et al. "Large-Scale Acoustic Automobile Fault Detection: Diagnosing Engines Through Sound." *KDD*, 2022.